

PHYSICS PROBLEM (START)

⚙️ TOOLS

Symbolic math engine

Numerical compute

Equation catalog

Plotting

Expression registry

- Creation of keys: LLM cannot create new keys only tools can
- Gated access: Tools only accept keys from registry or equation catalog
- Limited writes: LLM can only create simple variables (x , E , V_0) and substitute into existing keys
- Tracked entries: Each key stores: symbolic expression (hidden), description, and plot label

Reasoning Agent
Claude Sonnet 4.5

✓ CHECKER

Expression sourcing

Normalization

Physics sanity checks

END

**Pass expr
keys**

**Final
answer**

**Tool
result**

**Pass/
reject**